

Pape DIAWARA

Python Software Engineer - Data & Research Platforms

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Summary

Software Engineer with experience building scalable data systems, backend services, and research-oriented platforms in production environments. Strong focus on performance, reliability, and large-scale data processing. Experienced in developing tools and frameworks used by analytics and research teams.

Profiles

GITHUB
[capinho](#)

Programming

Skills

- **Programming Languages:** Python (FastAPI, Django), JavaScript/TypeScript (Node.js, React)
- **Data & Performance:** Pandas, NumPy, Polars, AsyncIO, ETL pipelines, vectorized processing, profiling, memory optimization, LLMs / automated data classification
- **Distributed Systems & Cloud:** AWS (EC2, S3, EKS, RDS), Kubernetes, Docker, Helm, Redis
- **Databases:** PostgreSQL, SQL, relational databases, query optimization
- **Software Engineering:** pytest, TDD, CI/CD, GitLab CI/CD, code reviews, testing, monitoring

Education

EPITA - School of Engineering and Computer Science, PARIS Computer Engineering Degree • Master
Computer Science
<https://www.epita.fr/en/homepage/>

Experience

BNP Paribas Asset Management Paris
Data / Software Engineer February 2025 - Present

- Designed and developed Python backend services and reusable data components supporting investment analytics and automated reporting workflows.
- Built asynchronous and distributed data-ingestion workflows integrating multiple internal and external data sources, with pagination, concurrency control, retry mechanisms and caching.
- Processed and consolidated portfolio, performance, risk, benchmark and product datasets using Pandas, NumPy and Polars.
- Improved processing performance and memory efficiency through vectorized transformations, caching and optimization of high-volume data workflows.
- Designed reusable APIs and transformation layers used by internal analytics and business applications.
- Deployed and operated services using Docker, Kubernetes, Helm, Redis, PostgreSQL and GitLab CI/CD.
- Improved production reliability through monitoring, retry strategies, data-quality controls and root-cause investigation.
- Contributed to code quality and maintainability through code reviews, automated testing, engineering standards and technical guidance.

SEGULA TECHNOLOGIE Paris
Software / Data Engineer 2022-11-29 - 2024-10-30

- Built Python microservices and APIs with Django and FastAPI for predictive-maintenance, Web3 and R&D platforms.
- Designed data-modeling frameworks and analytics backends to support algorithmic experimentation and model analysis.
- Built web-scraping and data-ingestion pipelines using Python, Selenium and BeautifulSoup to collect and structure external data for analytics and search workflows.
- Implemented asynchronous processing and caching layers to improve throughput and latency.
- Developed semantic-search and LLM-based components for intelligent data retrieval and classification.
- Contributed to internal research and visualization tools used by R&D teams for prototyping and experimentation.

GAWL, Paris 2022-09-01 - 2022-11-24
Fullstack / Data Engineer

- Built a Python/Django analytics platform backed by PostgreSQL.
- Designed ETL and aggregation pipelines for high-volume operational data.
- Improved backend performance through SQL optimization and caching.
- Implemented data validation, monitoring and alerting mechanisms.

Freelance 2019 - 2022
Freelance Developer

- Designed and delivered Python backend applications, internal tools and automation systems for SMEs.
- Managed projects end-to-end from requirements and architecture to deployment, maintenance and performance improvements.

SEDIMA GROUP 2017-11-01 - 2018-08-31
Fullstack Engineer

- Built internal platforms with data workflows for order, payment, and inventory management
- Developed data-driven applications with automation and reporting features
- Improved internal processes through workflow automation and structured data handling

Emc2 Group 2017-07-03 - 2017-08-31
Backend Software Engineer

- Developed backend features for an internal alumni platform using Python and Django.
- Designed relational data models, user-management workflows and administrative features.
- Contributed to backend testing, debugging and maintenance.